



8, 13, 21, 34, 55, 89 ...?

Counting spirals

Fibonacci numbers are common in flower-heads. If you look closely at the coneflower below, you'll see that the small florets are arranged in spirals running **clockwise** and **anticlockwise**. The number of spirals in each direction is a Fibonacci number. In this case, there are exactly 21 clockwise spirals and 34 anticlockwise spirals.



*clockwise
spirals*



*anticlockwise
spirals*

WHY?

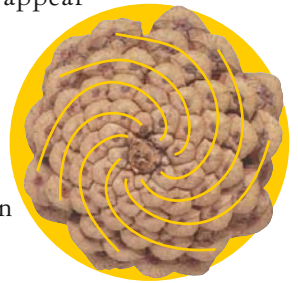
Why do Fibonacci numbers keep cropping up in nature? In the case of rabbits, they don't. Rabbits actually have more than two babies per litter and breed much more quickly than in Fibonacci's famous puzzle. But the numbers do crop up a lot in plants. They happen because they provide the best way for packing seeds, petals, or leaves into a limited space without large gaps or awkward overlaps.



FAQ

Cauliflowers and cones

It's not just flowers that contain **Fibonacci spirals**. You can see the same patterns in **pine cones**, **pineapple skin**, **broccoli florets**, and **cauliflowers**. Fibonacci numbers also appear in leaves, branches, and stalks. Plants often produce branches in a winding pattern as they grow. If you count upwards from a low branch to the next branch directly above it, you'll often find you've counted a Fibonacci number of branches.



Musical numbers

One octave on a piano keyboard is made up of 13 keys: 8 **white keys** and 5 **black keys**, which are split into groups of 3 and 2. Funnily enough, all of these are Fibonacci numbers. It's another amazing Fibonacci coincidence!

